Public Infrastructure Investment and Corporate Taxations

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Motivation and questions

- Infrastructure spending and its effect on output have become central issue in recent policy discussions.
  - The Infrastructure Investment and Jobs Act ($>$ $2 trillion dollars)
- Question (1) Is the infrastructure spending beneficial?
  - Infrastructure spending multipliers
  - Elasticity of substitution between private and public capital
- Question (2) How does financing matter for multipliers?
  - corporate taxation vs. lump-sum payment

This paper

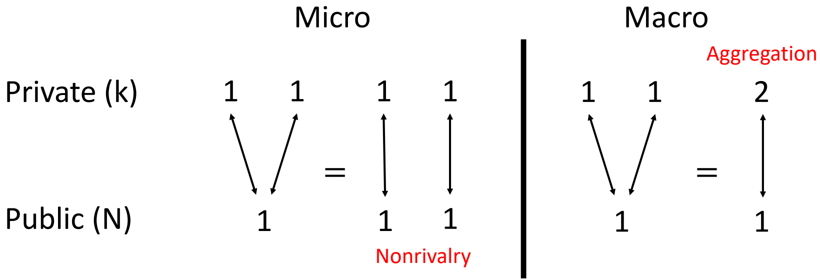
- estimates a heterogeneous firm general equilibrium model to quantify the fiscal multiplier of infrastructure expenditure.
- analyzes the role of corporate taxation on the fiscal multipliers.
- extends an existing method in a novel way to efficiently estimate a heterogeneous agent general equilibrium model.

Preview of result

- We obtain the state-level elasticity between public and private capital which is 0.44.
- Using our heterogeneous firm model, we estimate the firm-level elasticity between public and private capital which is 1.18.
  - Private capital is a gross substitute for public capital at the firm level, while it is a gross complement at the state level.
- We provide a theory to reconcile the gap between the micro and the macro elasticity. (firm-level vs. state-level)
- Given our estimate, the short-run infrastructure fiscal multiplier is greater than unity. (1.04 for output)
- Heavier corporate tax dampens the infrastructure fiscal multiplier as firms are less willing to invest.

Simple example

Consider a CRS-CES production function $f(k, N; \lambda)$, where λ is the elasticity between k and N. Suppose we have $k_1 = 1$, $k_2 = 1$ and $N = 1$.



$$y = 2f(1,1,\lambda^{micro}) \qquad y = f(2,1,\lambda^{macro})$$
$$= 2f(1,0.5,\lambda^{macro})$$

$$\Longrightarrow f(1,1,\lambda^{micro}) = f(1,0.5,\lambda^{macro})$$

$$\Longrightarrow \lambda^{micro} \neq \lambda^{macro}$$

Simple theory

Consider a CRS-CES production function $f = f(k, N; \lambda, z)$.

Proposition

Suppose we are given with the micro-level data set (k_1, k_2, y_1, y_2, N) *s.t.*

$$\exists i \in \{1,2\},\ s.t.\ k_i < N, \quad N \leq k_1 + k_2, \quad \frac{y_1}{k_1} = \frac{y_2}{k_2}$$

Suppose the micro-level estimates (z, λ) *and the macro-level estimate* ξ *are exactly identified by fitting the data into the production function as follows:*

$$f(k_1, N; \lambda, z) = y_1$$
$$f(k_2, N; \lambda, 1) = y_2$$
$$f(k_1 + k_2, N; \xi, 1) = y_1 + y_2$$

Then, if the micro-level input elasticity satisfies $\lambda \geq 1$*, then the macro-level input elasticity satisfies* $\xi < 1$*.*

Empirical estimates of input elasticity

- We use state-level data on the real public highway infrastructure investment from Bennett et al. (2020).
- We estimate the elasticity of substitution between private and public capital given a CES production technology.

$$\ln\left(\frac{Y_{it}}{Y_{it-1}}\right) = c + (1-a)\ln\left(\frac{L_{it}}{L_{it-1}}\right) + \frac{a}{\psi}\ln\left[\frac{bK_{it}^{\psi} + (1-b)N_{it}^{\psi}}{bK_{it-1}^{\psi} + (1-b)N_{it-1}^{\psi}}\right] + (\epsilon_{it} - \epsilon_{it-1}).$$

- The elasticity $1/(1-\psi)$ is estimated to be 0.4406.
- The state-level variations indicate the complementarity between private and public capital.

Model

Roadmap

Model

1. Overview of the model
2. Production technology
3. Firm-level investment
4. Government
5. Recursive formulation (Firm)
6. Household
7. Stationary recursive general equilibrium

Overview of the model economy

Firms

Heterogeneous firms across two regions with different infrastructure and productivity.

Convex adj. cost + fixed cost. Details

Government

Infrastructure expenditure, lump-sum subsidy, and public employment.

Income tax, corporate tax, and bond issuance.

Household

A representative household consumes, works, and saves (claim for all firms).

Competitive market

Production technology

In the beginning of a period, a firm i is given capital stock k_{it}, its location j, idiosyncratic productivity z_{it}. A firm i in state j operates using a CES production function:

$$z_{it}x_{jt}A_t f(k_{it}, l_{it}, \mathcal{N}_{jt}) = z_{it}x_{jt}A_t \left(\theta(k_{it})^{\frac{\lambda-1}{\lambda}} + (1-\theta)\mathcal{N}_{jt}^{\frac{\lambda-1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}\alpha} l_{it}^{\gamma}, \quad j \in \{P, G\}$$

where

- z_{it} is an idiosyncratic productivity: discretized AR(1) process.
- x_{jt} is a state-specific productivity: j follows two-state Markov process.
- A_t is an aggregate productivity: fixed at unity (stationary equilibrium).
- λ is one of the key parameters.

Government

The government budget balance satisfies

$$\begin{aligned}\mathcal{G}_t &= \tau^h(w_t l_t + D_t) + \int \tau^c \pi(z_{it}, k_{it}, j; A_t, \mathcal{N}_t, w_t, r_t) d\Phi_t + \frac{B_{t+1}}{1+r_t^B} - B_t \\ &= \mathcal{F}_t + w_t \mathcal{E}_t + T_t\end{aligned}$$

- Financing: Income Tax + Corporate Tax + Bond Issuance
- Spending: Infra. Spending + Employment + Lumpsum Subsidy

The public capital stock $N_{A,t}$ evolves in the following law of motion:

$$\begin{aligned}\mathcal{N}_{A,t+1+s} &= \mathcal{N}_{A,t+s}(1-\delta_{\mathcal{N}}) + \mathcal{F}_{t+1} - \frac{\mu}{2}\left(\frac{\mathcal{F}_{t+1}}{\mathcal{N}_{A,t+s}}\right)^2 \mathcal{N}_{A,t+s} \\ \mathcal{N}_{jt} &= \zeta_j \mathcal{N}_{A,t} \quad j \in \{P, G\}\end{aligned}$$

where j refers to either poor or good states. ($\zeta_P + \zeta_G = 1$)

Recursive formulation (Firm)

$$J(k,z,j;S) = [\pi(k,z,j;S)(1-\tau^c) + (1-\delta)k](1-\tau^h) + \int_0^{\overline{\xi}} \max\left\{R^*(k,z,j,\xi;S), R^c(k,z,j;S)\right\} dG_\xi(\xi)$$

$$R^*(k,z,j,\xi;S) = \max_{k'} \quad [-k' - c(k,k') - w(S)\xi](1-\tau^h) + \mathbb{E}m(S,S')J(k',z',j';S')$$

$$R^c(k,z,j;S) = \max_{k^c-(1-\delta)k\in\Omega(k)} \quad [-k^c - c(k,k^c)](1-\tau^h) + \mathbb{E}m(S,S')J(k^c,z',j';S')$$

(Convex adjustment cost) $c(k,k') := \left(\mu^I/2\right)\left((k'-(1-\delta)k)/k\right)^2 k$

(Fixed cost) $w(S)\xi, \quad \xi \sim_{iid} Unif[0,\overline{\xi}]$

(Constrained investment) $I^c \in \Omega(k) := [-k\nu, k\nu] \quad (\nu < \delta)$

(Idiosyncratic productivity) $z' = G_z(z)$ (AR(1) process)

(Stochastic discount factor) $m(S,S') = \beta\left(C(S)/C(S')\right)$

(Aggregate states) $S = \{A, \Phi\}$

(Aggregate law of motion) $\Phi' := H(S)$

Stationary recursive competitive equilibrium

$$\text{[Capital Market]} \quad \int \underbrace{\mathbb{E}J(z', k'(z,k); \mathcal{S}) d\Phi}_{\text{Capital Demand}} = \underbrace{a'(a; \mathcal{S})}_{\text{Capital Supply}}$$

$$\text{[Labor Market]} \quad \int \underbrace{\left(n(z,k,j;\mathcal{S}) + \left(\frac{\xi^*(z,k,j;\mathcal{S})}{2} \right) \right) d\Phi}_{\text{Private Labor Demand}} + \underbrace{\mathcal{E}}_{\text{Public Labor Demand}} = \underbrace{L(a;\mathcal{S})}_{\text{Labor Supply}}$$

$$\text{[Aggregate Dividends]} \quad D(\mathcal{S}) = \int \left(\pi(z,k,j;\mathcal{S})(1-\tau^c(k)) - I^*(z,k,j;\mathcal{S}) - InvAdjCost \right) d\Phi$$

$$\text{[Ex-dividend Portfolio Value]} \quad P(\mathcal{S}) = \int J(z,k,j;\mathcal{S}) d\Phi - D(\mathcal{S})$$

$$\text{[Government Budget]} \quad \mathcal{G}(\mathcal{S}) = \tau^h(w(\mathcal{S})L(a^{Eq};\mathcal{S}) + D(\mathcal{S})) + \int \tau^c(k)\pi(z,k,j;\mathcal{S})d\Phi + \frac{B}{1+r^B(\mathcal{S})} - B$$

$$\text{[Infrastructure Investment]} \quad \mathcal{F}(\mathcal{S}) = \varphi(\mathcal{G}(\mathcal{S}) - w(\mathcal{S})\mathcal{E})$$

$$\text{[Lump-sum Subsidy]} \quad \mathcal{T}(\mathcal{S}) = (1-\varphi)(\mathcal{G}(\mathcal{S}) - w(\mathcal{S})\mathcal{E})$$

$$\text{[Infrastructure]} \quad \mathcal{N}_A = \frac{1+\sqrt{1-2\mu\delta_{\mathcal{N}}}}{2\delta_N}\mathcal{F}(\mathcal{S}), \quad \mathcal{N}_j = \zeta_j \mathcal{N}_A \quad j \in \{P, G\}$$

Estimation

Estimation

- Some parameters are externally calibrated. e.g. corporate tax rate 0.27, Frisch elasticity 4 (Ramey, 2020).
- Challenge: The market clearing prices have to be solved for each candidate value for the model parameters.
- Our method: The limited-information Bayesian method extended to augment general equilibrium conditions as moments to match. Moments
  - We additionally use the following market clearing conditions:

$$\begin{bmatrix} p - 1/c(\mathcal{S}) \\ w - \eta L(\mathcal{S})^{\frac{1}{\chi}} c(\mathcal{S})/(1-\tau^h) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

  - We estimate market clearing prices simultaneously with the model parameters.

Estimation

- The elasticity of substitution parameter is identified from the high region's k portion.
  - An exogenous variation in the public capital stock leads to variation in the private capital stock as firms endogenously change their investment.
- The firm-level elasticity is estimated to be 1.1848. Details
- As external validation, we compute the state-level elasticity from our model (0.35) and compare it to the empirical counterpart (0.44). Details
- Our model bridges the gap between the firm-level and the state-level estimates. The non-rivalry of the infrastructure generates the complementarity between the state-level private and public capital.

Quantitative analysis

Measurement of fiscal multiplier

Following Ramey (2020), we measure the fiscal multiplier as follows:

$$Fiscal\ Multiplier = \frac{\sum_{t=1}^{T} Present\ Value\ of\ \Delta x_t}{\sum_{t=1}^{T} Present\ value\ of\ \Delta G_t}$$

where Δx_t is the deviation at period t of the equilibrium allocation of interest from the steady-state level; ΔG_t is the fiscal spending shock at period t.[1]

- In the short run, we assume $T = 2$.
- In the long run, we assume $T = 5$.

The fiscal spending shock ΔG at $t = 1$ affects the public capital stock in the following way:

$$\mathcal{N}_{A,t+1} = \mathcal{N}_{A,t}(1 - \delta_{\mathcal{N}}) + \mathcal{F}_{t+1} - \frac{\mu}{2}\left(\frac{\mathcal{F}_{t+1}}{\mathcal{N}_{A,t}}\right)^2 \mathcal{N}_{A,t}$$

$$\mathcal{N}_{jt} = \zeta_j \mathcal{N}_{A,t} \quad j \in \{P, G\}$$

$$F_t = \begin{cases} F^{ss} + \Delta G & \text{if } t = 1 \\ F^{ss} & \text{otherwise} \end{cases}$$

[1] The shock is deviation from the steady-state level.

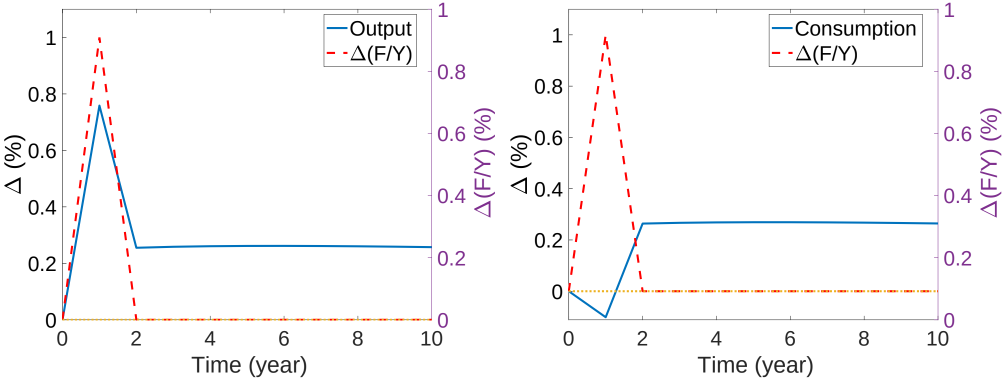
Fiscal multipliers and corporate taxation

- Fiscal multiplier is slighty beyond unity in the baseline.
- Fiscal multiplier decreases in the corporate tax rate.
- Investment is crowded-in under the low corporate taxation:
  - General equilibriume effect $<$ Complementarity effect

Table: Fiscal multipliers

Fiscal multipliers	Low Corp. Tax	Baseline	High Corp. Tax
Output			
Short-run	1.1637	1.0416	0.9210
Long-run	2.0758	1.8439	1.6149
Investment			
Short-run	0.0224	-0.0942	-0.2094
Long-run	0.0671	-0.0617	-0.1889

Impulse responses to the infrastructure spending shock

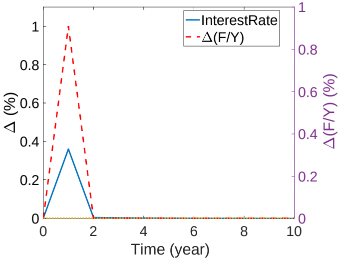


(a) Output (b) Consumption

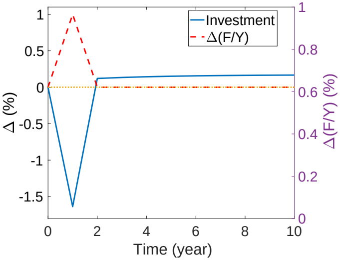
Figure: Impulse responses of output and consumption

Fiscal multipliers and GE effect

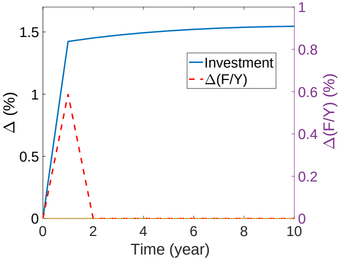
- Upon the fiscal spending shock, the interest rate jumps up.
- This is due to drop in the contemporaneous consumption (prev. slide)
- Investment drops substantially due to the rise in the interest rate.
- Without GE effect, infrastructure investment triggers a greater private investment.



(a) Interest rate



(b) GE



(c) PE

Figure: Impulse responses of interest rate and investment in GE and in PE

Fiscal multipliers and the elasticity between private and public capital

- The elasticity between private and public capital significantly affect the fiscal multipliers.
- The elasticity's effect on marginal benefit of private investment (direct and indirect).

Table: Fiscal multipliers with the different elasticities of substitutions

Fiscal multipliers	High ($\lambda = 3$)	Estimated ($\lambda = 1.18$)	Low ($\lambda = 0.5$)
Output			
Short-run	0.6129	1.0416	1.3296
Long-run	0.7734	1.8439	2.6087
Investment			
Short-run	-0.2257	-0.0942	0.0532
Long-run	-0.3087	-0.0617	0.2242

Conclusion

- This paper quantifies the infrastructure investment multipliers.
- With a heterogeneous firm model, we provide a micro-foundation for the multipliers with firms' investment decisions.
- We find that private capital is a gross substitute for public capital at the firm level, while it is a gross complement of public capital at the state level.
- We show that heavier corporate taxation leads to dampened fiscal multipliers.
- A potential topic for future research is to study the optimal corporate taxation given an infrastructure spending plan.

Model fit

Target moment	Data	Model
lumpy investment portion	0.140	0.139
mean(i/k)	0.100	0.100
standard deviation(i/k)	0.160	0.160
private-to-public capital ratio	0.750	0.798
high region's k portion	0.830	0.870
high region's y portion	0.849	0.984
government spending to output ratio	0.155	0.154
employment rate	0.330	0.344

Back

Firm-level investment

Total adjustment costs $=$ Fixed cost $+$ convex adjustment cost

- Fixed cost: $w\xi$
  - Incurs only if $I \notin \Omega(k) := [-\nu k, \nu k]$ $(\nu < \delta)$
  - $\xi \sim_{iid} Unif([0, \overline{\xi}])$
  - As in Khan and Thomas (2008), there exists a threshold rule for the extensive-margin adjustment: Adjust if $\xi^*(k, z; S) > \xi$.
  - The cost is regarded as an overhead labor cost.
- Convex adjustment cost: $c(k, I) = \frac{\mu}{2}\left(\frac{I}{k}\right)^2 k$
  - Essential component to match the empirical elasticity of the aggregate investment (Zwick and Mahon, 2017; Koby and Wolf, 2020; Lee, 2022). Back

Estimation

Parameter	Description	Posterior Distribution Mean	Posterior Distribution 95% Interval	Uniform Prior [Min, Max]
$\bar{\xi}$	fixed cost upper bound	0.5188	[0.5112,0.5230]	[0.001,1.900]
μ	convex adjustment cost	3.1244	[3.1175,3.1348]	[0.200,3.500]
ν	constrained investment region	0.0406	[0.0405,0.0407]	[0.001,0.080]
θ	private capital share	0.6674	[0.6658,0.6684]	[0.500,0.999]
λ	elasticity of substitution	1.1848	[1.1800,1.1900]	[0.300,2.500]
x	productivity of high infra. region	2.0637	[2.0421,2.0801]	[0.500,2.500]
G	government spending level	0.1030	[0.1014,0.1068]	[0.010,0.400]
η	labor disutility	2.8445	[2.8311,2.8597]	[2.100,3.500]

- The elasticity of substitution parameter is identified from the high region's k portion.
  - An exogenous variation in the public capital stock leads to variation in the private capital stock as firms endogenously change their investment.
- The firm-level elasticity of substitution supports the Cobb-Douglass production function as a reasonable specification. Back

Estimation

- As external validation, we compute the state-level elasticity from our model and compare it to the empirical counterpart (0.44).
- State-level aggregation: We fix the micro-level estimates except for the elasticity λ and spatial productivity heterogeneity x_1.
  - We estimate (λ, x_1) under the state-level production models.

$$\begin{bmatrix} x_1 \left(\theta k_1^{\frac{\lambda-1}{\lambda}} + (1-\theta) N_1^{\frac{\lambda-1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}\alpha} l_1^{\gamma} \\ \left(\theta k_2^{\frac{\lambda-1}{\lambda}} + (1-\theta) N_2^{\frac{\lambda-1}{\lambda}}\right)^{\frac{\lambda}{\lambda-1}\alpha} l_2^{\gamma} \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

  - $(x_1, \lambda) = (1.7664, 0.3493)$.
- Our model bridges the gap between the firm-level and the state-level estimates. The non-rivalry of the infrastructure generates the complementarity between the state-level private and public capital. Back

Fiscal multipliers and time-to-build

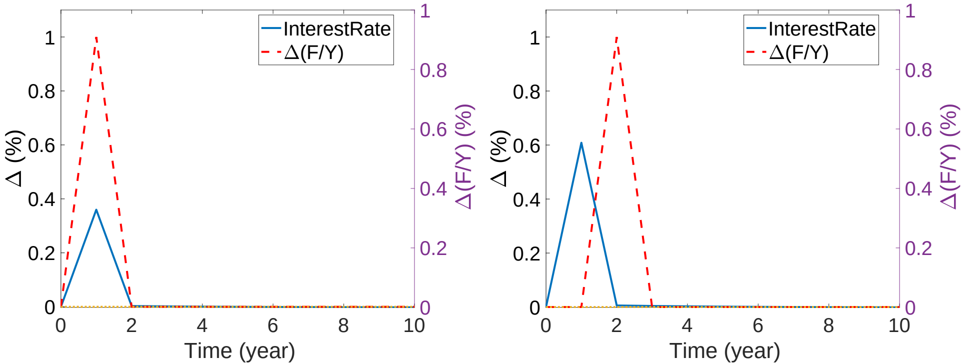
- Time to build of one year leads to a lower fiscal multiplier.
- This is dominantly driven by strong GE effect.
  - In the PE, fiscal multiplier is greater with T2B.
  - This is due to greater increase (drop) in the contemporaneous interest rate (consumption) when T2B>0. (next slide)

Fiscal multipliers	Y	Y & T2B	I	I & T2B
GE				
Short-run	1.0416	0.9213	-0.0942	-0.1627
Long-run	1.8439	1.6881	-0.0617	-0.1333
PE				
Short-run	1.8582	2.0301	0.1892	0.2897
Long-run	5.0314	5.4015	0.4769	0.5858

Table: Fiscal multipliers across the states under the time to build of one year

General equilibrium effect and time to build

- A greater drop in contemporaneous consumption leads to a sharper rise in interest rate.



(a) Baseline (b) T2B

Figure: Impulse response of interest rate